

Lesson 5

Managing Server Hardware

Topic 5A

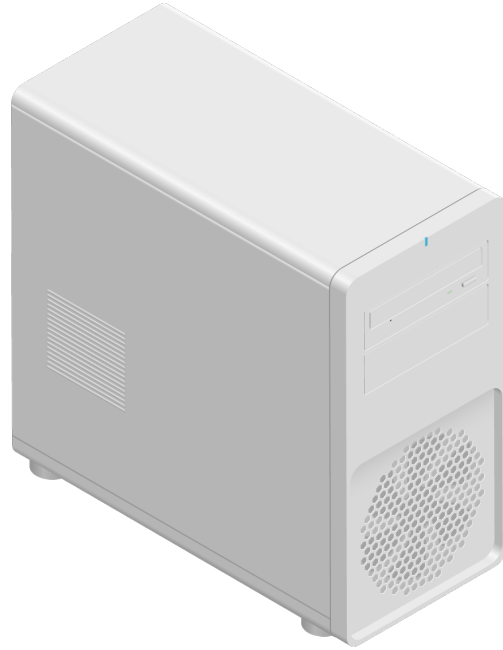
Manage the Physical Server

Exam Objectives Covered

- 1.1 Given a scenario, install physical hardware

Server Chassis Types

- Tower form factor



*Server in the tower form factor, very similar in size and appearance to a typical workstation.
(Image © 123RF.com)*

Server Chassis Types (slide 2)

- Blade form factor



Server in the blade form factor, installed within a chassis. (Image © 123RF.com)

Server Chassis Types (slide 3)

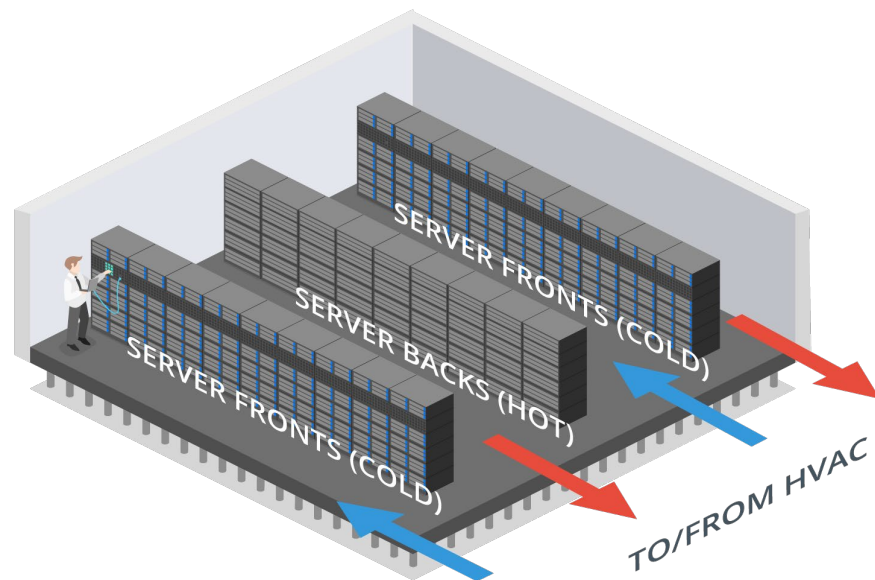
- Rack form factor



Server rack holding multiple components. (Image © 123RF.com)

Mounting Server and Network Devices in Racks

- Rail kits
- Rack layout for cooling
- Environmental sensors
- Air conditioning
- Hot and cold aisles in the server room



Server orientation with hot and cold airflow. (Image © 123RF.com)

Mounting Server and Network Devices in Racks (slide 2)

- Safety



Hydraulic trolley, one type of equipment used for safely lifting servers to the appropriate height in a rack. (Image © 123RF.com)

Mounting Server and Network Devices in Racks (slide 3)

- Power distribution Unit (PDU)
- Keyboard video mouse (KVM) placement



Rack-mounted KVM. (Image © 123RF.com)

Power Cabling

- Power connections
- NEMA 5-15P/R



*NEMA 5-15P (a three-prong plug) and NEMA 5-15R (a three-prong receptacle).
(Image © 123RF.com)*

Power Cabling (slide 2)

- IEC C13/C14



IEC C13 AND C14 cable terminals. (Image by Andrii Zhezhera © 123RF.com)

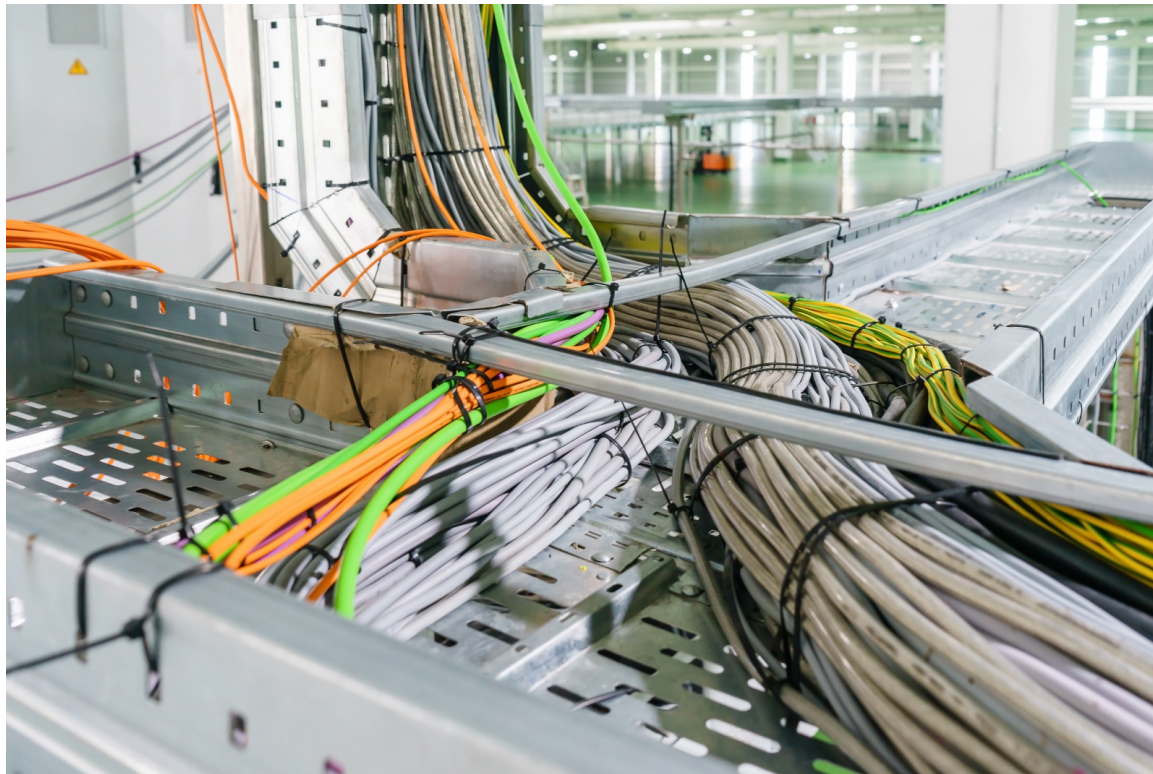
Power Cabling (slide 3)

- Cable management



Horizontal cable managers can be used for power and for network cables. (Image by Kjetil Kolbjornsrud © 123RF.com)

Power Cabling (slide 4)



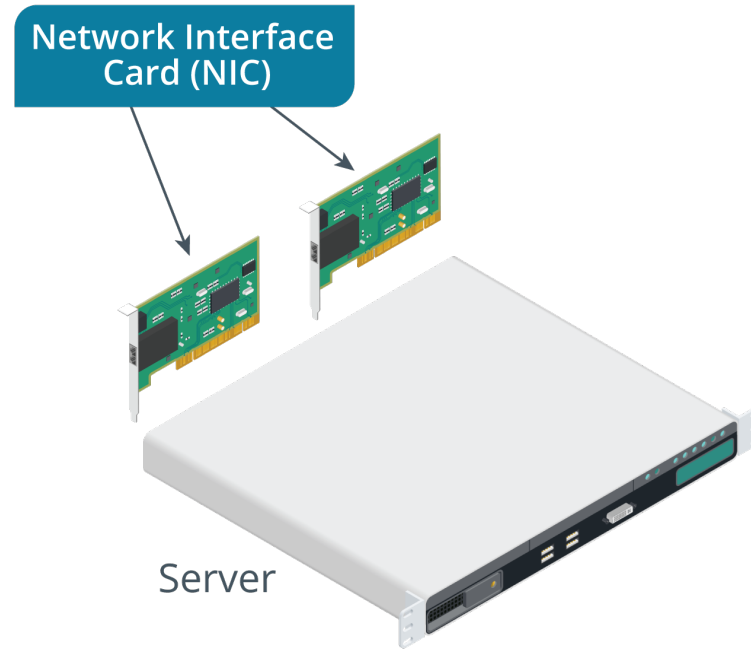
Overhead cable tray supporting bundles of cable. (Image by Pisit Khambubpha © 123RF.com)

Network Cabling

- Connectivity
- Redundancy

Redundancy

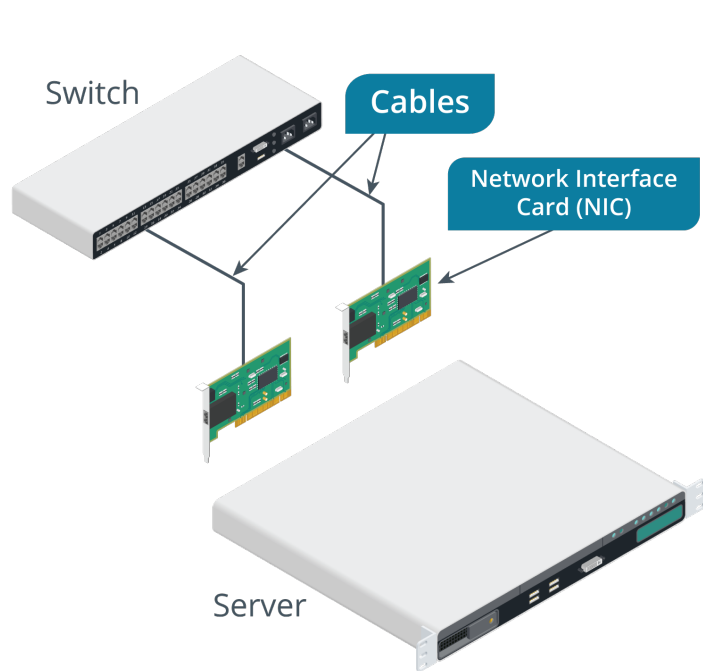
- Networking
- Server



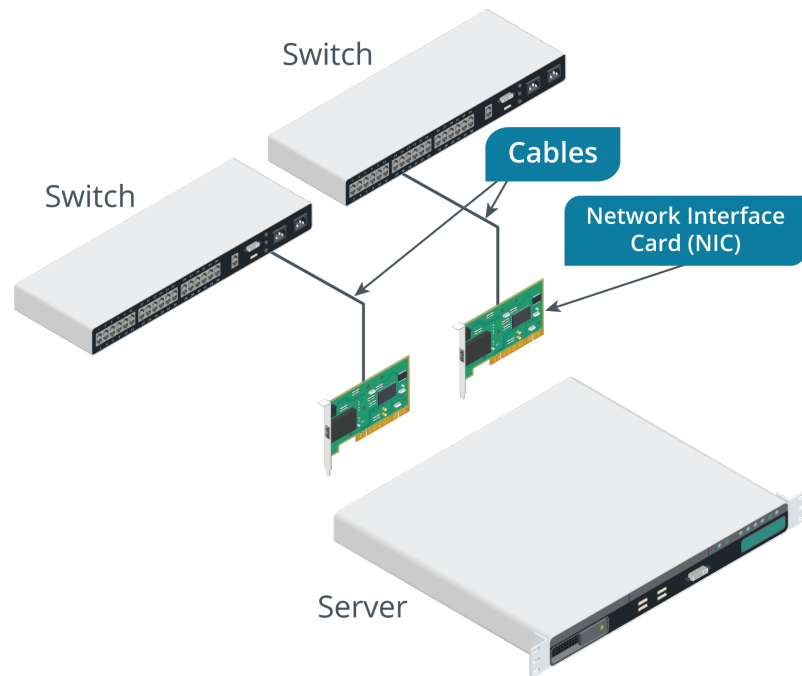
Server with redundant NICs. (Images © 123RF.com)

Redundancy (slide 2)

- Media and switch



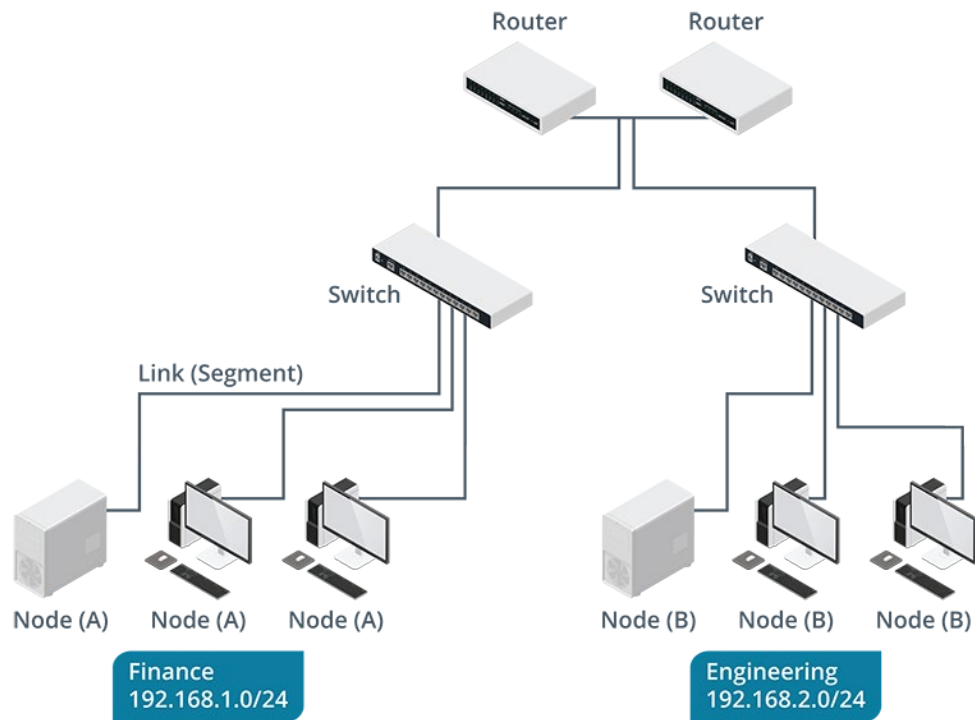
Server with redundant NICs plugged into one switch. (Images © 123RF.com)



Server with redundant NICs plugged into redundant switches. (Images © 123RF.com)

Redundancy (slide 3)

- Router



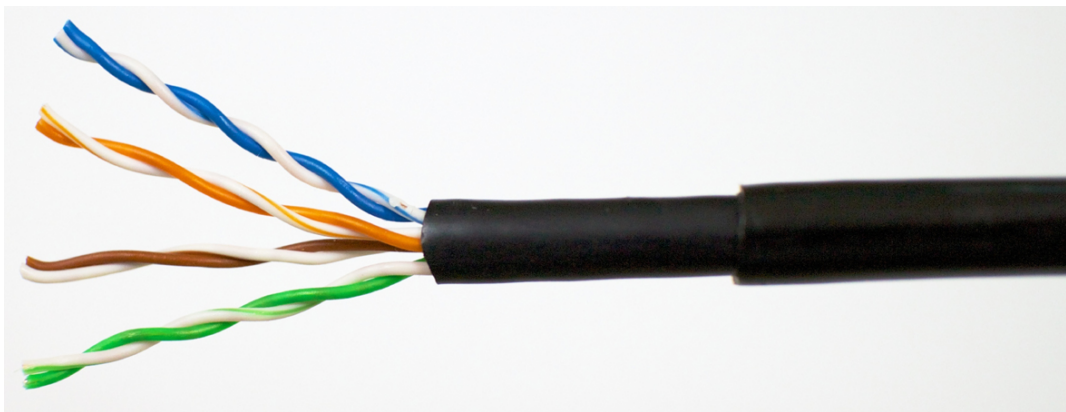
Redundant routers between network segments. (Images © 123RF.com)

Transmission Media Types

- Copper
- Fiber optic

Transmission Media Types (slide 2)

- Twisted pair cable
 - Unshielded Twisted Pair (UTP)



Twisted pair cable with four pairs, twisted at different rates. (Image by Thuansak Srilao © 123RF.com)

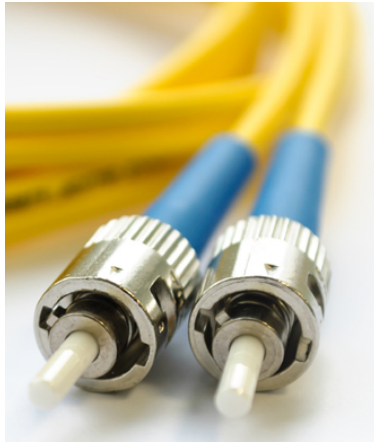
- Shielded Twisted Pair (STP)

Transmission Media Types (slide 3)

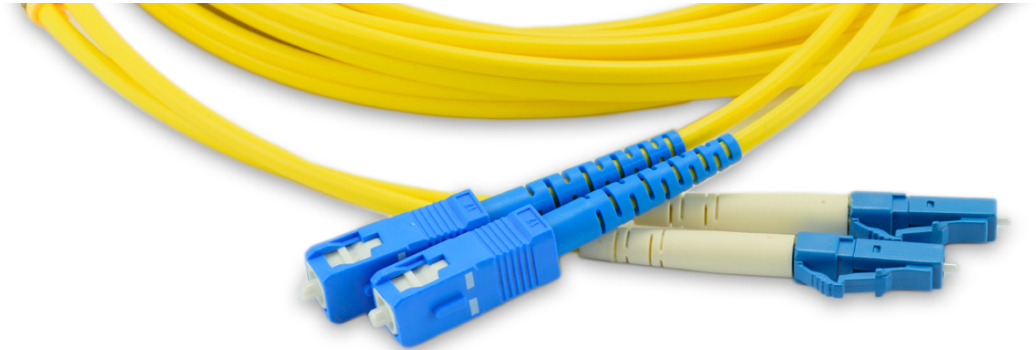
- RJ-45 connectors
- Cat standards
- Gigabit
- 10GbE

Transmission Media Types (slide 4)

- Fiber optic cable
 - Single mode and multimode
 - Connector types



Two single-mode ST connectors. (Image by Aleh Datskevich © 123RF.com)



Fiber optics single-mode patch cord with different connectors, SC to LC. (Image by Yanawut Suntornkij © 123RF.com)

Transmission Media Types (slide 5)

- Transceivers and small form factor pluggable (SFP), SFP+, quad SFP

Review Activity: Physical Server Management



Review Activity

Topic 5B

Administer the Server and Storage

Exam Objectives Covered

- 1.1 Given a scenario, install physical hardware
- 1.3 Given a scenario, perform server hardware maintenance

Local Hardware Administration and Out-of-Band Management



RJ-45 Ethernet port



RJ-45 ports for out-of-band management. (Image © 123RF.com)

Local Hardware Administration and Out-of-Band Management

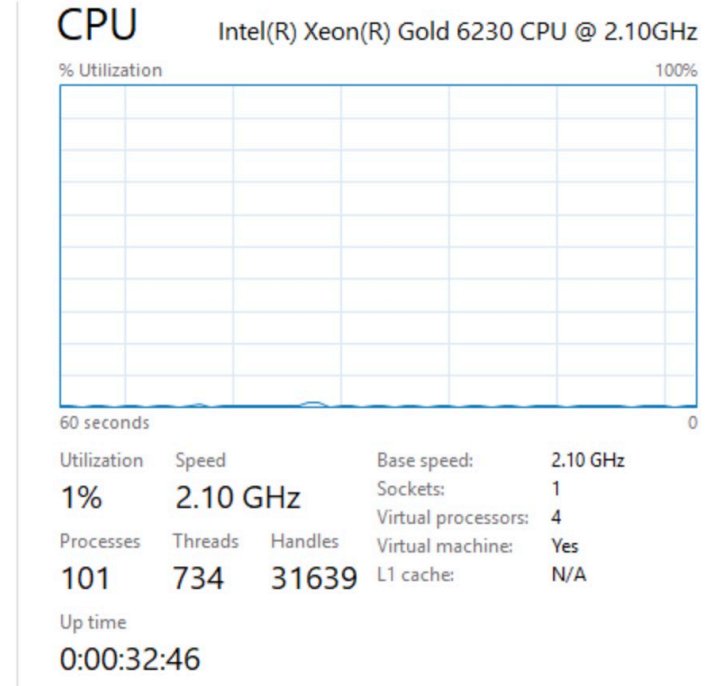
- Crash cart



Crash cart in a server room. (Image by Aleksei Gorodenkov © 123RF.com)

Manage Server Components

- Hardware Compatibility List (HCL)
- Central Processing Unit (CPU)



Windows Task Manager with four virtual processors displayed at lower right. (Screenshot courtesy of Microsoft.)

Manage Server Components (slide 2)

- GPU
- Memory
- OS memory requirements
- Expansion
- Server motherboard buses
- Expansion bus types
- Interface types



RS-232 COM port. (Image by Yuri Mineav © 123RF.com)

Storage Drives

- Storage disk maintenance
 - chkdsk
 - fsck

Hot-swappable hardware

- Cages
- Cards
- Power supplies and fans



Hot-swappable HDD tray. (Image by kubais © 123RF.com)

Review Activity: Administration and Storage



Review Activity

Topic 5C

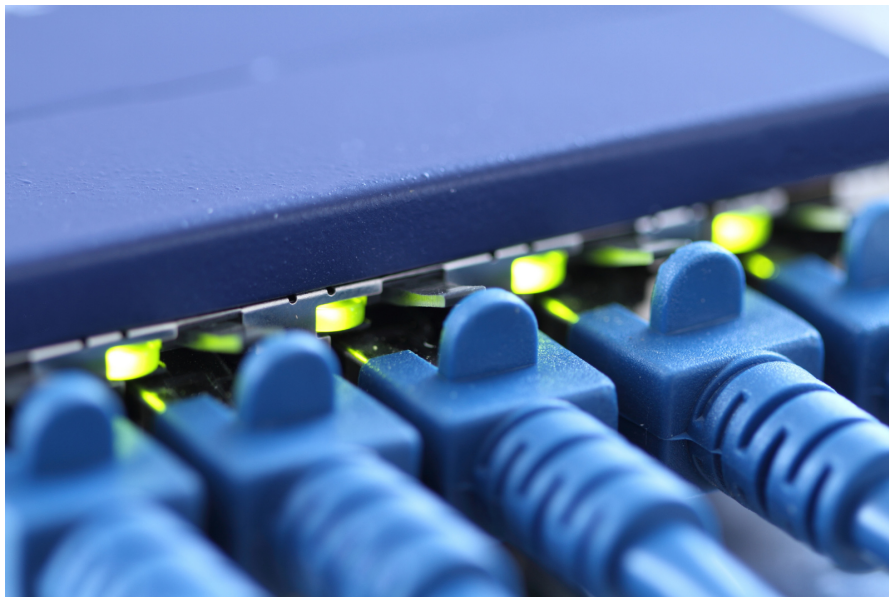
Troubleshoot Server Hardware

Exam Objectives Covered

- 4.2 Given a scenario, troubleshoot common hardware failures

Indicators of Common Hardware Problems

- Predictive failure analysis
- Visual, auditory, and olfactory indicators
- System crashes and lockups
- Memory failures
- CMOS battery failure



LED indicators on a switch. (Image by sbmh765 © 123RF.com)

Causes of Common Hardware Problems

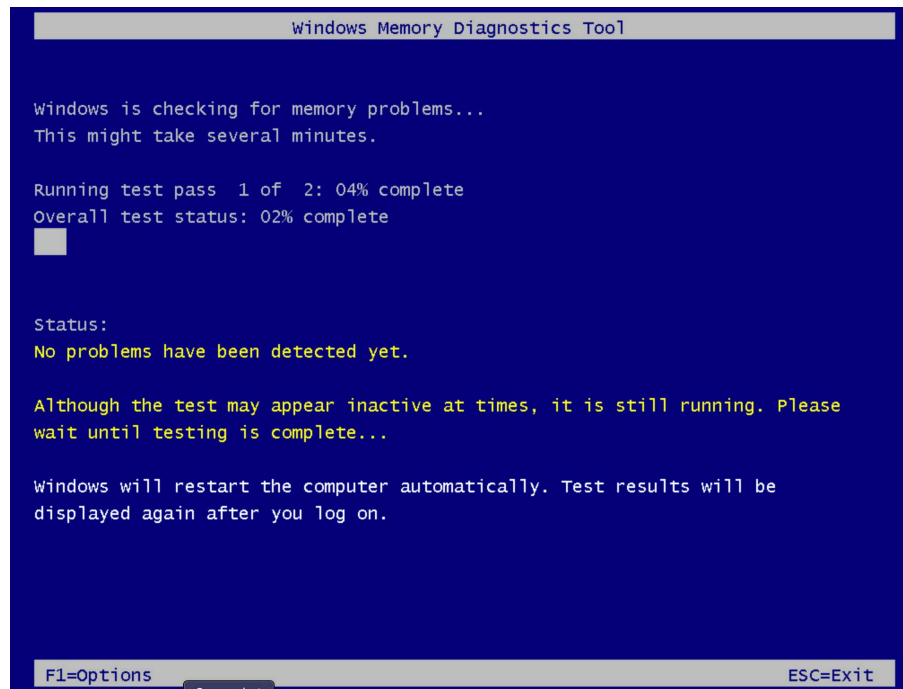
- Technical
- Environmental
- Configuration



Environmental sensors can alert administrators. (Image by primagefactory © 123RF.com)

Tools and Techniques for Troubleshooting Hardware

- Preparing a workspace for troubleshooting
- Physical hardware troubleshooting
- Firmware and diagnostic tools



Windows Memory Diagnostic Tool. (Screenshot courtesy of Microsoft.)

Tools and Techniques for Troubleshooting Hardware (slide 2)

- Log files

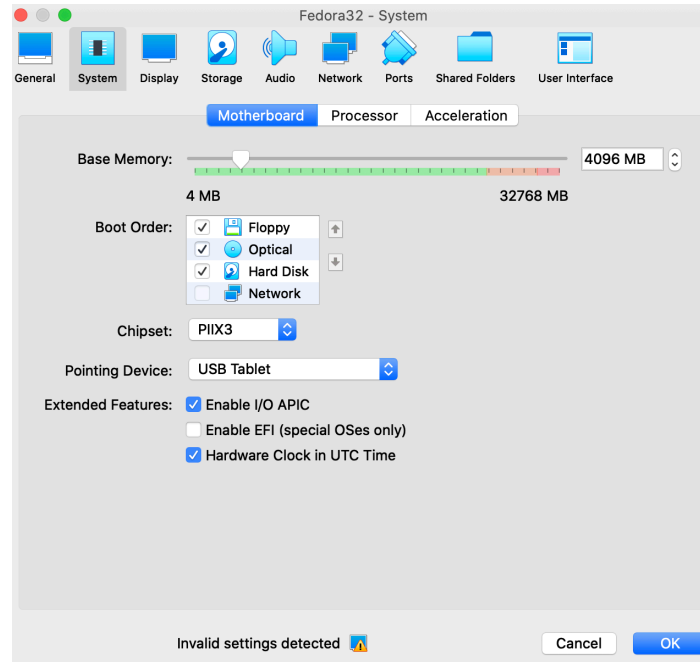
```
[root@localhost ~]# dmesg | grep network
[ 0.876171] drop_monitor: Initializing network drop monitor service
[ 4.396153] SELinux: policy capability network_peer_controls=1
[ 4.396154] SELinux: policy capability always_check_network=0
[root@localhost ~]#
```

Gather network information with dmesg.

- Power-on self-test (POST) codes
- LCD panel readouts

Tools and Techniques for Troubleshooting Hardware (slide 3)

- Misallocation of resources for vms



VirtualBox VM with a Base Memory allocation of 4 GB.

Review Activity: Server Hardware



Review Activity

Lesson 5

Summary

